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SUPERSEDING
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March 17, 1959

FEDERAL SPECIFICATION
TAPE, PRESSURE-SENSITIVE ADHESIVE
(CELLOPHANE AND CELLULOSE ACETATE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for cellophane and cellulose acetate adhesive tapes with or without dispensers used for paper mending, attaching, shielding, and temporary color marking.

1.2 Classification. The tapes covered by this specification shall be of the following types and classes, as specified (see 6.2):

Type I - Cellophane
Class A - Transparent
Class B - Colored

Type II - Cellulose acetate, transparent,
Class A - Matte finish
Class B - Glossy finish

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

PPP-T-680 - Tape, Pressure Sensitive Adhesive: Packaging and Packing of.

Federal Standard:

Fed. Test Method Std. No. 147 - Tapes, Packaging, Adhesive and Gummed; Methods of Inspection, Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

3. REQUIREMENTS

3.1 Materials. Type I tape shall be cellophane film. Type II, class A tape shall be cellulose acetate film having a matte finish, and capable of being written on with pen, pencil and other commonly used marking devices. Type II, class B tape shall be cellulose acetate film having a glossy finish.

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3.2 Adhesive. The adhesive shall be pressure-sensitive, and shall require no moisture, heat or other special preparation prior to application of the tape. The adhesive shall completely cover one side of the tape backing (see 6.4).

3.3 Colors. Type I, class B tape shall be furnished in the following colors as specified (see 6.2), and shall be a good commercial match to the color chart available for comparison at each of the General Services Administration Regional Business Centers.

Red	Light Green
White	Light Blue
Black	Orange Yellow
Medium Green	Orange
Yellow	Dark Blue
Medium Blue	

3.4 Transparency. Type I, class A tape and type II, classes A and B tapes shall be sufficiently transparent to allow easy reading of five point type when one layer of the tape is pressed down over the type. In application, type II, class A tape shall not cause distortion when photographed or reproduced by other standard methods.

3.5 Rolls. The tape shall be uniformly and evenly wound, adhesive side in, on cores of sufficient rigidity to prevent distortion of the rolls under normal conditions of transportation, storage, and use. No more than three splices will be permitted in any roll. All splices shall be such that they will not separate when the roll is unwound in test or application. The lengths and widths of rolls, and inner diameters of cores shall be as indicated in table I. A width tolerance of ± 1/32 inch shall be allowed. Utility style rolls shall be complete with dispensers (see 3.6) as indicated in table I.

3.6 Dispensers. Dispensers shall be made of metal or other suitably strong and rigid material, having a serrated cutting edge sufficiently sharp to permit the tape to be cut cleanly without causing the tape to wrinkle or tear unevenly, and having all other edges rounded to prevent physical injury to the hands. Dispensers shall have sufficient space between the edge of the roll and the cutter for the insertion of the index finger.

TABLE I.

Type	and class	Dispenser	Inside diameter of cores (inches)	Width and length of tape (inches)
I	A	Yes	1	1/2 x 400
I	A	Yes	1	3/4 x 300
II	A	Yes	1	1/2 x 400
II	A	Yes	1	3/4 x 300
I, II	A, B	No	1	1/2 x 1296
I, II	A, B	No	1	3/4 x 1296
I, II	A, B	No	3	* x 2592

* The widths for 2592-inch rolls shall be 1/2-, 3/4-, 1-, 1-1/2-, 2-inch, or other standard commercial widths, as specified (see 6.2).

3.7 Properties. The tape shall comply with the requirements specified in table II and in 3.7.1 through 3.7.3.

TABLE II.

	Type I	Type II	Test
Adhesion, oz/in width, min.			
Initial	20	12	4.4.2
After accelerated aging by light	20	12	4.4.2
After accelerated aging	20	-	4.4.2
Tensile, lbs./in. width, min.	20	12	4.4.2
Thickness, In., Max.	.0035	.0035	4.4.2
Unwind, oz/in width, max.	40	30	4.4.2

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3.7.1 Accelerated aging. After accelerated aging (see 4.4.2), types I and II tapes shall be capable of being unwound at a rate of 20-30 inches per second without adhesive offsetting, splitting, stringiness, nor tearing, breaking or delamination of the backing. Adhesion requirements of table II shall also be met.

3.7.2 Shrinkage. Type II tape, when tested as described in 4.4.2, shall not shrink more than ten percent.

3.7.3 Mending stability. Type II tape, when tested as described in 4.4.2, shall not cause the sheets of bond paper to adhere together.

3.8 Workmanship. Workmanship shall be first class throughout. The tape shall be free of defects which detract from its appearance or impair its serviceability. The adhesive shall be a smooth uniform coating, covering the entire area of one side of the tape. Edges shall be straight, true, and unbroken.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 End item examination. Sampling and examination of the item shall be as specified in section 7 of Fed. Test Method Std. No. 147.

4.3 End item testing. Sampling and examination for end item testing shall be as specified in section 7 of Fed. Test Method Std. No. 147 at an Acceptable Quality Level (AQL) of 4 percent.

4.4 Tests. Tests for acceptance of individual lots shall consist of tests for all requirements specified in section 3.

4.4.1 Specimen. The specimen shall be selected and prepared as described in Method 4.1 of Fed. Test Method Std. No. 147.

4.4.2 Test methods. The following tests shall be conducted as described in Fed. Test Method Std. No. 147:

	Method
Adhesion	- 10.2
Tensile	- 30.2
Thickness	- 36.2
Unwind	- 80.2
Accelerated Aging (Heat and Humidity Method)	- 60.2
Shrinkage	- 38.2
Mending Stability	98.1

The accelerated aging by light test shall be conducted as described in 4.4.2.1.

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4.4.2.1 Accelerated aging by light. Three 1/2-inch specimens of the tape 8 inches long and 5 specimens 12 inches long shall be used for this test. Each 8 inch strip shall be adhered to the bond paper described in method 98.1 of Fed. Test Method Std. No. 147 by rolling twice with a roller using only the force of the roller's weight. Each 12-inch strip shall be attached to one side of a 4 inch by 8 inch glass plate, starting at the center and placing 4 inches of the tape in the lengthwise direction of the plate, adhesive side against the glass, and then looping the tape back on itself to give an 8 inch length having its adhesive side directly exposed to the light. The free end of each test specimen shall be taped in place. Test specimens on bond paper and on glass plates shall be placed on the slowly rotating platform of a dustproof cabinet, equipped with a type R. S. Sunlamp (General Electric), or equivalent, in such a manner that the longitudinal axis of the paper or glass plate shall be along a radius of the turntable and the corners of one end of the paper or glass plate shall touch the edge of the turntable. The distance from the lowest point of the glass envelope of the lamp to the center of the turntable shall be 30 inches. The turntable shall have a maximum diameter of 20 inches. The intensity of the lamp shall be controlled by a variable auto-transformer to 2250 microwatts per square centimeter of U. V. between 300 and 400 nanometers (excluding infrared). The measurement shall be with the base of the meter 12 inches from the lowest point of the lamp envelope and at the point of highest intensity. The meter shall be a model J-221, available from Ultra-Violet Products, Inc., or equivalent.

The voltage to the auto-transformer shall be controlled within limits of plus or minus 1 percent by a constant-voltage transformer. The temperature at the surface of the turntable shall be maintained at 120° ± 5°F. The lamp shall have been in operation for a total of 100 hours before being used for this test, and for a minimum of 3 hours immediately before any test. The specimens adhered to bond paper and those on glass plates shall be exposed for the specified time to the light, then removed and conditioned according to method 1.1 in Fed. Test Method Std. No. 147. Then the adhesion value of the test specimens on glass plates shall be tested as specified (see 4.4.2), except that the adhesion value of the 2 inches of each specimen of tape nearest the center of the glass plate shall be reported as the adhesion value after accelerated aging by light. All specimens shall be examined for evidence of adhesive deterioration. The adhesive shall be considered unsatisfactory if the adhesive of any of the specimens adhered to bond paper penetrates the paper or causes partial transparency of it; if the adhesion of any of the specimens having the adhesive directly exposed does not comply with the requirements for adhesion after accelerated aging by light (see 3.7); if, in either case, the adhesive of any one of the specimens becomes stringy or separates from the tape; or if the tape has become excessively discolored when placed against a white background and compared with tape which has not been exposed to this test. Type I tape shall be tested for 30 hours. Type II tape shall be tested for 120 hours.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing. Packaging and packing shall be in accordance with the requirements of PPP-T-680. The level of packaging and packing shall be A, B or C, as specified in the contract or order (see 6.2).

5.2 Marking. Containers of tape shall be marked in accordance with the requirements of PPP-T-680.

6. NOTES

6.1 Intended uses.

6.1.1 Type I. Class A tape is intended for temporary applications such as paper mending, attaching, covering, and general office use. Class B tape is intended for temporary applications such as identification, charting, decorating and holding.

6.1.2 Type II. Class A tape is a dimensionally stable long-aging tape designed for long-lasting mending of paper, documents, books, maps, etc. Because the tape is invisible when applied to paper, such things as documents, blue prints, maps, etc., may be copied by standard reproducing methods. In addition, it will accept typing and writing with pencil, pen, and other common marking devices. Class B tape is intended for long-lasting application such as light holding, covering, and shielding. Because it is a dimensionally stable long-aging tape, it is ideally suited for protecting such things as labels, etc.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Type and class of tape required (see 1.2).
(c) Width and length of tape required (see table I).
(d) Color, as applicable (see 3.3).
(e) Selection of applicable levels of packaging and packing (see 5.1).
(f) Marking required (see 5.2).

6.3 Storage. The tape should be stored in the original container in a cool location. It should not be stored in close proximity to steam pipes, radiators, or other sources of heat, or in a location where it may be exposed to direct rays of the sun.

6.4 A condition sometimes exists in cellophane and cellulose acetate tape, wherein the adhesive surface appears to be pockmarked. These pockmarks result from small amounts of air entrapped between the layers of tape during winding and in no way affect the quality of the tape. This condition should not be considered evidence of lack of uniformity of adhesive coating.

MILITARY INTERESTS:

DOD has waived coordination on revisions and amendments to this Federal Specification until further notice.

CIVIL AGENCY COORDINATING ACTIVITIES:

DOT-ACO
HEW-FDA
NASA-LRC
TVA-TVA

PREPARING ACTIVITY: GSA-FSS

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